Problem Set #1: Foundations of Reinforcement Learning

1 The Monty Hall Problem

(Anderson) The Monty Hall problem is a famous math problem loosely based on a television quiz show hosted by Monty Hall. You face three closed doors. There is a big prize behind one door, but nothing behind the other two. You are asked to pick one door out of the three, but you cannot yet see what is behind it. Monty opens one of the two other doors to reveal that there is nothing behind it. Then he offers you one chance to switch your choice. Should you switch?

(a)

Suppose you will not switch. What is the probability that you win? (5 points)

P=1/3

(b)

Suppose you will switch. Monty shows a door without a prize. What is the probability that you win? (5 points)

if you originally chose an empty door, you will win after switch. But if you chose the prize door, you will lose. So the probability to win equals to the probability you chose an empty door originally.

So P=2/3

2 Committee Selection

(Anderson) Ten men and five women are meeting in a conference room. Four people are chosen at random from the fifteen to form a committee.

(a)

What is the probability that the committee consists of two men and two women? (5 points)

P=$\frac{C_{10}^2 C_5^2}{C_{15}^4} = \frac{30}{91} = 32.967\%$

(b)

Among the fifteen is a couple, Bob and Jane. What is the probability that Bob and Jane both end up on the committee? (5 points)

P=$\frac{C_{13}^2}{C_{15}^4} = \frac{2}{35} = 5.714\%$

3 Uniform Distribution on a Triangle

(Anderson) Pick a uniformly chosen random point inside the triangle with vertices $(0, 0)$, $(3, 0)$, and $(0, 3)$.

(a)

What is the probability that the distance of this point to the y-axis is less than 1? (5 points)

P=$\frac{\frac{1}{2}(3+2)\times 1}{\frac{1}{2}\times 3\times 3} = \frac{5}{9} = 55.56\%$

(b)

What is the probability that the distance of this point to the origin is more than 1? (5 points)

P=$1 - \frac{\frac{1}{4}\times\pi\times 1}{\frac{1}{2}\times 3\times 3} = 1 - \frac{\pi}{18} = 82.547\%$

4 Randomly Chosen Die

(Anderson) An urn contains one 6-sided die, two 8-sided dice, three 10-sided dice, and four 20-sided dice. One die is chosen at random and then rolled.

(a)

What is the probability that the roll gave a five? (10 points)

P=$\frac{1}{10} \times \frac{1}{6} + \frac{2}{10} \times \frac{1}{8} + \frac{3}{10} \times \frac{1}{10} + \frac{4}{10} \times \frac{1}{20} = \frac{11}{120} = 9.17\%$

(b)

What is the probability that the die rolled was the 20-sided die, given that the outcome of the roll was seven? (10 points)

P$(D_{20}|7) = \frac{P(7|D_{20})\times P(D_{20})}{P(7)} = \frac{\frac{1}{20}\times\frac{4}{10}}{\frac{2}{10}\times\frac{1}{8}+\frac{3}{10}\times\frac{1}{10}+\frac{4}{10}\times\frac{1}{20}} = \frac{4}{15} = 26.67\%$

5 Defective Phones

(Anderson) Two factories I and II produce phones for brand ABC. Factory I produces 60% of all ABC phones, and factory II produces 40%. Ten percent of phones from factory I are defective, and twenty percent produced from factory II are defective. If you purchase a brand ABC phone, and assume that it is not defective, what is the probability that it comes from factory II? (20 points)

$P(II|ND) = \frac{P(ND|II)\times P(II)}{P(ND)} = \frac{0.8\times 0.4}{0.6\times 0.9+0.4\times 0.8} = \frac{16}{43} = 37.21\%$

6 Discrete Random Variable

(Anderson) Let X have possible values $\{1, 2, 3, 4, 5\}$ and probability mass function

x	1	2	3	4	5
$p_X(x)$	$\frac{1}{7}$	$\frac{1}{14}$	$\frac{3}{14}$	$\frac{2}{7}$	$\frac{2}{7}$

(a)

Calculate $P(X \le 3)$. (5 points)

$P(X \le 3) = \frac{2+1+3}{14} = \frac{3}{7}$

(b)

Calculate $P(X < 3)$. (5 points)

$P(X < 3) = \frac{2+1}{14} = \frac{3}{14}$

7 Expectation of a Quadratic Function

Suppose that the random variable X has expected value $\mathbb{E}[X] = 3$ and variance $\mathrm{Var}(X) = 4$. Compute $\mathbb{E}[(2X+3)^2]$. (10 points)

$\because Var(X) = 4,\ E[(X - E(X))^2] = 4$

$\therefore E[X^2 + 9 - 6X] = 4,$ so we get $E[X^2] = 13$

$E[(2X+3)^2] = E[4X^2 + 9 + 12X] = 4E[X^2] + 9 + 12E[X] = 97$

8 Normal Distribution

Suppose that a random variable X has the normal distribution with mean 95 and standard deviation 11. Find the probability $P(X \ge 100)$. (Hint: Use the cumulative distribution function of the standard normal random variable Z and its table of values.) (10 points)

$X \sim N(\mu = 95, \sigma = 11)$

$Z = \frac{X-\mu}{\sigma} = \frac{X-95}{11} \sim N(0,1)$

$P(X \geq 100) = P(Z \geq \frac{5}{11}) = 1 - \phi(0.45455) = 0.3247 = 32.47\%$